University of Arkansas STAT6393V - Week 1

Week 1 Report: Optimizing Linear Regression using Distribution Theory

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Introduction

Previously, we wrote our linear regression model as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$,
where $\epsilon_i \sim N(0, \sigma^2)$.
Turn out, using the properties of the normal distribution, an equivalent specification is: $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$.
So instead of using $\sum \epsilon_i^2$ as the loss function, we can also use the negative log likelihood of the normal distribution as the loss function.
Consider the following data generation process:

```r
set.seed(0)
n <- 1000
x <- rnorm(n = n, mean = 0, sd = 2)
b0 <- -2; b1 <- 7.7
y <- rnorm(n, b0 + b1 * x, sd = 6)
```

The likelihood of the normal distribution for y_i, ..., y_n (where n = 1000) is:
$L(\beta|y) = \frac{1}{(6\sqrt{2\pi})^n} \exp\left\{-\frac{1}{2 \cdot 6^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right\}$
where $\beta = (\beta_0, \beta_1)$
Estimate β using the following steps:

1. Write down an expression of the negative log-likelihood (nll) - $-l(\beta|y) = -logL(\beta|y)$
2. Use the `optim` function to minimize the nll and estimate β.
3. Confirm your answer by using the `dnorm` function instead of manually calculating the nll.
4. [BONUS] In the previous parts, we fixed $\sigma = 6$. Can you also try estimating it from the data as well? You will need to estimate $\theta = (\beta_0, \beta_1, \sigma)$ where the first components are unbounded but the last one is strictly positive.

Answer 1

- The likelihood of the normal Distribution:

$$L(\beta|y) = \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{-\frac{1}{2\cdot\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2\right\}$$

- The expression of the negative log-likelihood (nll):
  $nll = -l(\beta|y) = -\log L(\beta|y) \Rightarrow -l(\beta|y) = -\log\left(\frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{-\frac{1}{2\cdot\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2\right\}\right)$
- Using some properties of logarithms, we obtain the following simplified expression for the nll:
  $nll = -l(\beta \mid y) = +n\log(\sigma\sqrt{2\pi}) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2$.
- In our case, since $\sigma = 6$, and $n = 1000$, the expression becomes:
  $nll = 1000\log(6\sqrt{2\pi}) + \frac{1}{72}\sum_{i=1}^{1000}(y_i - \beta_0 - \beta_1 x_i)^2$.

Answer 2

- Write the nll function to be minimized by `optim` and then minimize it:

```
nll_loss <- function(theta){
  b0 = theta[1]
  b1 = theta[2]
  nll = n * log(6*sqrt(2*pi)) + sum((y - b0 - b1*x)^2)/72
  return(nll)
}
```

- Minimizing the nll function:

```
optim(c(0, 0), nll_loss)
```

```
$par
[1] -2.151580  7.660794

$value
[1] 3244.403

$counts
function gradient
      85       NA

$convergence
[1] 0

$message
NULL
```

Answer 3

- Confirming the answer using the `dnorm` function instead of manually calculating the nll:

```
ll <- function(theta){
  b0 = theta[1]
  b1 = theta[2]
  # Sum of the logs of the density at each point in the y vector
  llik = sum(dnorm(y, mean = b0 + b1 * x, log = T))
  # Negative log-likelihood to be minimized
  return(-llik)
}
```

- Minimizing the nll function:

```
optim(c(0, 0), ll)
```

```
$par
[1] -2.151580  7.660794

$value
[1] 20132.32

$counts
function gradient
      85       NA

$convergence
[1] 0

$message
NULL
```

Bonus Question

```
nll_Sig <- function(theta){
  b_0 = theta[1]
  b_1 = theta[2]
  sigma = theta[3]
  nll = n * log(sigma*sqrt(2*pi)) + sum((y - b_0 - b_1*x)^2)/(2*(sigma)^2)
  return(nll)
}
```

- Minimizing the nll function:

```r
optim(c(0, 0, 1),
      nll_Sig,
      method = "L-BFGS-B",
      lower = c(-Inf, -Inf, 1e-10),
      upper = c(Inf, Inf, Inf))
```

```
$par
[1] -2.149962  7.660663  6.199288

$value
[1] 3243.316

$counts
function gradient
      18       18

$convergence
[1] 0

$message
[1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
```